

Mode A: Learning ID + Opt. Control

Online Parameter/Intent
Identification (Learning)

State/Params

MPC / QP Trajectory
Optimization (Model-based)

Mode B: Opt. Control + Learning Adapt

Force Compliance &
Stability Control (Model-based)

Interaction

Preference Learning &
Weight Adaptation (Learning)

Mode C: Learning Plan + Opt. Control

Policy-based Path
Planning (Learning)

Commands

Safe Execution &
Action Shielding (Model-based)

Mode D: Hierarchical Hybrid

High-level Strategy
Selection (Learning)

Reference

Mid/Low-level Motion
& Safety Control (Model-based)